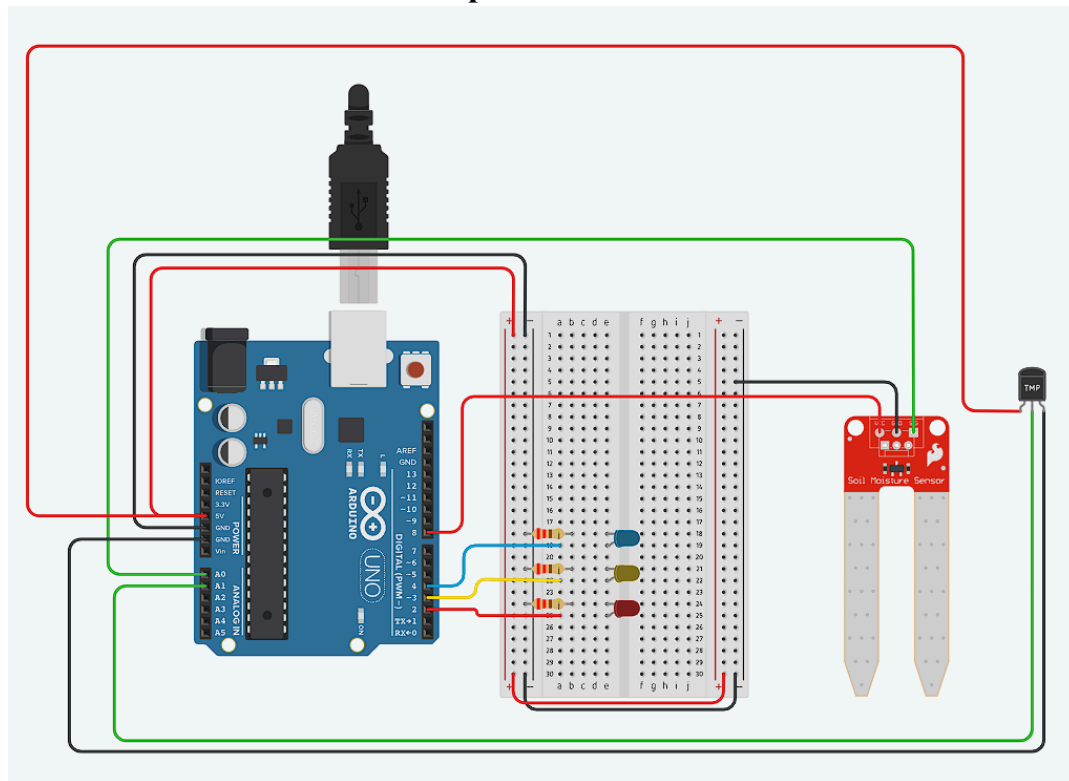


Indian Institute of Information Technology Vadodara,
International Campus Diu



Group: ICD-8

Arduino with soil sensor and temperature sensor in tinkercad:



```
const int sensorPin = A0;
const int tempPin = A1;
const int sensorPower = 8;

const int LED1 = 2;
const int LED2 = 3;
const int LED3 = 4;

int sensorValue;
int tempValue;

int wet = 800;
int dry = 500;

const float referenceVoltage = 5.0;
const float resistorValue = 220.0;
const float ledForwardVoltage = 2.0;

void setup() {
    pinMode(LED1, OUTPUT);
```

```
pinMode(LED2, OUTPUT);
pinMode(LED3, OUTPUT);
pinMode(sensorPower, OUTPUT);

Serial.begin(9600);
Serial.println("Soil + Temperature Monitoring System");
}

void loop() {

    digitalWrite(sensorPower, HIGH);
    delay(10);

    sensorValue = analogRead(sensorPin);

    digitalWrite(sensorPower, LOW);

    tempValue = analogRead(tempPin);

    float soilVoltage = (sensorValue * referenceVoltage) / 1023.0;

    float tempVoltage = (tempValue * referenceVoltage) / 1023.0;

    float temperatureC = (tempVoltage - 0.5) * 100;

    float ledCurrent = (referenceVoltage - ledForwardVoltage) /
resistorValue;

    Serial.print("Soil ADC: ");
    Serial.print(sensorValue);

    Serial.print(" | Soil Voltage: ");
    Serial.print(soilVoltage);
    Serial.print(" V");

    Serial.print(" | Temp ADC: ");
    Serial.print(tempValue);

    Serial.print(" | Temp Voltage: ");
    Serial.print(tempVoltage);
    Serial.print(" V");
```

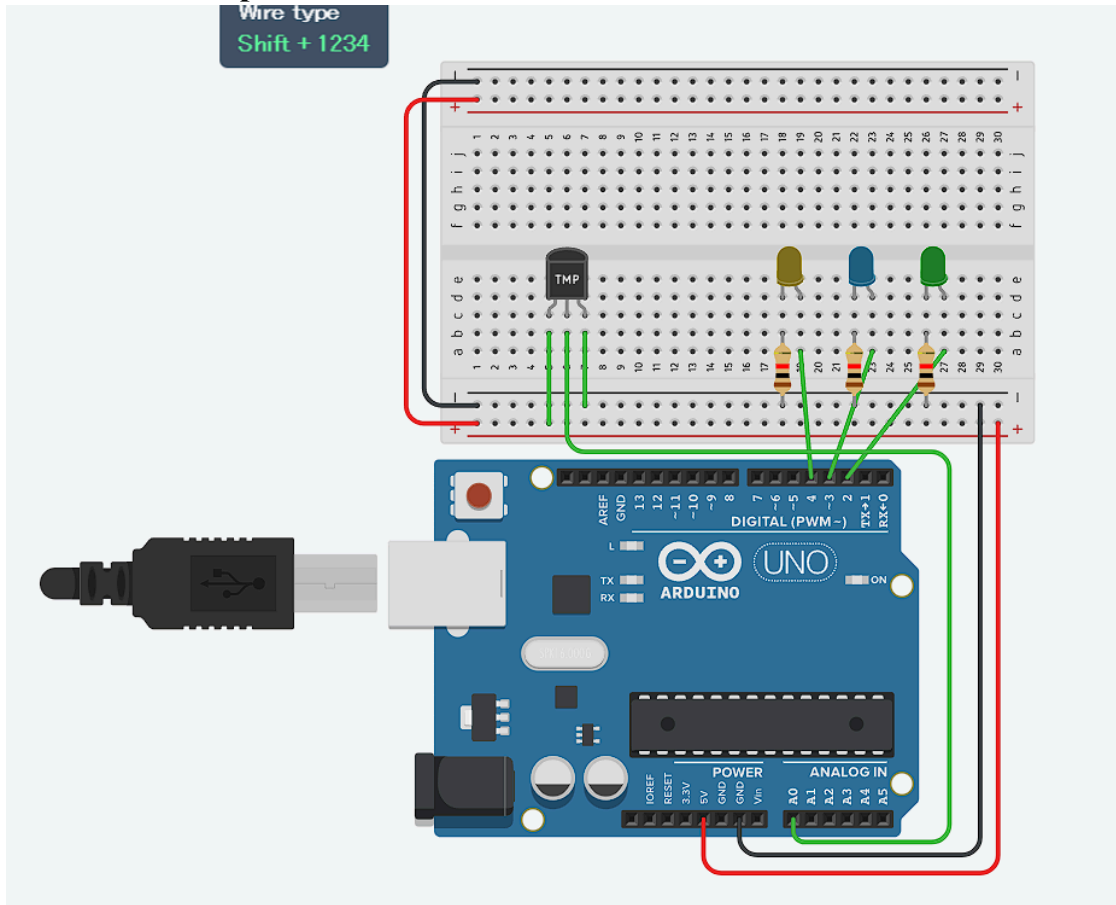
```
Serial.print(" | Temperature: ");
Serial.print(temperatureC);
Serial.print(" C");

Serial.print(" | LED Current: ");
Serial.print(ledCurrent * 1000);
Serial.println(" mA");

if (sensorValue < dry || temperatureC > 35) {
  Serial.println("Status: DRY Soil OR HIGH Temperature");
  digitalWrite(LED1, HIGH);
  digitalWrite(LED2, LOW);
  digitalWrite(LED3, LOW);
}
else if (sensorValue > wet && temperatureC <= 35) {
  Serial.println("Status: WET Soil & Normal Temp");
  digitalWrite(LED1, LOW);
  digitalWrite(LED2, LOW);
  digitalWrite(LED3, HIGH);
}
else {
  Serial.println("Status: MOIST Soil & Normal Temp");
  digitalWrite(LED1, LOW);
  digitalWrite(LED2, HIGH);
  digitalWrite(LED3, LOW);
}

Serial.println("-----");
delay(1000);
```

Arduino with temperature sensor in tinkercad:



```
int baselineTemp = 0;
int celsius = 0;
int fahrenheit = 0;

void setup()
{
  pinMode(A0, INPUT);
  Serial.begin(9600);

  pinMode(2, OUTPUT);
  pinMode(3, OUTPUT);
  pinMode(4, OUTPUT);
}

void loop()
{
  baselineTemp = 40;

  celsius = map((analogRead(A0) - 20) * 3.04, 0, 1023, -40, 125);
```

```
fahrenheit = ((celsius * 9) / 5 + 32);
Serial.print(celsius);
Serial.print(" C, ");
Serial.print(fahrenheit);
Serial.println(" F");

if (celsius < baselineTemp) {
    digitalWrite(2, LOW);
    digitalWrite(3, LOW);
    digitalWrite(4, LOW);
}
if (celsius >= baselineTemp && celsius < baselineTemp + 10) {
    digitalWrite(2, HIGH);
    digitalWrite(3, LOW);
    digitalWrite(4, LOW);
}
if (celsius >= baselineTemp + 10 && celsius < baselineTemp + 20) {
    digitalWrite(2, LOW);
    digitalWrite(3, HIGH);
    digitalWrite(4, LOW);
}
if (celsius >= baselineTemp + 20 && celsius < baselineTemp + 30) {
    digitalWrite(2, LOW);
    digitalWrite(3, LOW);
    digitalWrite(4, HIGH);
}
if (celsius >= baselineTemp + 30) {
    digitalWrite(2, LOW);
    digitalWrite(3, LOW);
    digitalWrite(4, HIGH);
}
delay(1000);
}
```